The Bi-Fractional Neural Sampling (bFNS) Model

The bi-fractional neural sampling (bFNS) model is defined by a pair of coupled equations on a one-dimensional position x and auxiliary momentum p:

$$\begin{aligned} {}^{C}D_t^{\beta} x &= -\eta \nabla \tilde{V}_{\alpha} + \gamma p + \eta^{\frac{1}{\alpha}} \xi_{\alpha,\beta} \,, \\ \frac{dp}{dt} &= -\gamma \nabla \tilde{V}_{\alpha} \,, \end{aligned} \tag{1}$$

where $\alpha \in (1, 2]$ is the spatial fractional order, $\beta \in (0, 1]$ is the temporal fractional order, γ is the momentum coupling strength, and η is the noise intensity. Here D_t is the standard temporal derivative, ∇ is the regular spatial gradient, and $\xi_{\alpha,\beta}(t)$ is a linear fractional stable noise process with Lévy order α and self-similarity exponent $H = \frac{1}{2} + \frac{1}{\alpha} - \frac{\beta}{2}$.

The Caputo fractional temporal derivative ${}^{C}D_t^{\beta}$ of order β is defined as:

$${}^{C}D_t^{\beta}[f(t)] = \frac{1}{\Gamma(1-\beta)} \int_0^t \frac{f'(\tau)}{(t-\tau)^{\beta}} \, d\tau \,, \tag{2}$$

where $\Gamma(z) = \int_0^{\infty} e^{-u} u^{z-1} \, du$ is the gamma function.

The effective potential $\tilde{V}_{\alpha}(x)$ is defined through its gradient as:

$$\nabla \tilde{V}_{\alpha}(x) = \frac{-\nabla \left[\mathcal{D}_x^{\alpha-2} \pi \right]}{\pi} \,, \tag{3}$$

where $\pi(x)$ is the target distribution and $\mathcal{D}_x^{\alpha-2}$ is the Riesz fractional derivative of order $\alpha - 2$. When $\alpha = 2$, this reduces to the classical Langevin potential $V(x) = -\log \pi(x)$.

The Riesz fractional spatial derivative of order α is defined in the Fourier domain by:

$$\widehat{\mathcal{D}_x^{\alpha} f}(k) = -|k|^{\alpha} \hat{f}(k) \,. \tag{4}$$